Portland State UNIVERSITY

Agile & Adaptive Robotics Lab

Pressure and Force Dynamics in Artificial Muscle Actuators: A State-Space and Optimization-Based Approach

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Abstract

This study investigates the dynamic behavior of braided pneumatic actuators (BPAs) under bio-inspired pulse modulation control to enhance their biomimetic performance. Building upon prior research on pulse timing and force amplification, we develop a state-space model to characterize the relationship and set parameter values between valve actuation and system pressure, optimized using Particle Swarm Optimization (PSO). A nonlinear model is then used to correlate pressure with force output via trust-region reflective least squares. Experiments were conducted on Festo BPAs at fixed lengths of 600 mm, 310 mm, and 140 mm, across various pulse frequencies and durations. Results show that the PSO-optimized model accurately predicts pressure behavior, with average errors of 5–16% depending on muscle length. The pressure-to-force model yielded errors below 3%, effectively capturing the nonlinear force response. Our findings reveal that shorter BPAs respond more quickly but require higher pulse frequencies to maintain force, while longer BPAs exhibit higher peak forces and slower deflation, enabling smoother force output. These insights contribute to the development of more adaptable and efficient soft robotic systems that mimic animal biomechanics.

Background

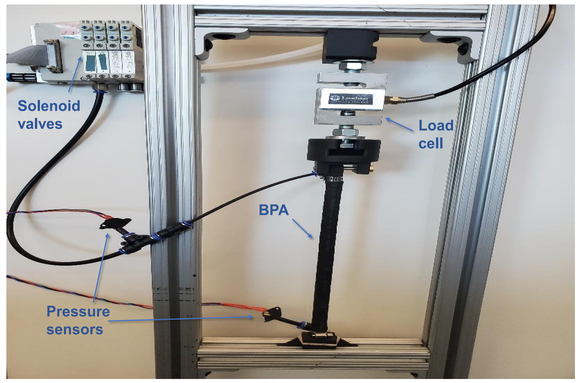
Braided Pneumatic Actuators (BPAs) have emerged as a promising alternative to conventional rigid actuators due to their inherent compliance, adaptability, and biomimetic potential. Originally developed in the 1950s as McKibben actuators for medical rehabilitation, these devices closely mimic the contraction dynamics of biological muscles by expanding radially and contracting longitudinally upon pressurization[1,2]. Specifically, the Festo fluidic muscle variant integrates an internal braid structure to enhance performance, allowing contractions with a final length that is up to 70–80% of their initial length[3].



Fig 1. A. A deflated Festo fluidic muscle BPA that consists of a braided rubber bladder, pressure sensors on each side, and end stops. **B.** Inflated Festo BPA. **C.** Festo fluidic muscle cross section.

Recent research has shown that applying bio-inspired pulse modulation (discrete bursts of air) can significantly enhance the biomimetic behavior of these actuators. This method not only replicates the neurological activation patterns of biological muscles but also demonstrates unique nonlinear phenomena such as force amplification through pulse timing[4,5]. Despite these advances, a comprehensive understanding of how pulse frequencies and actuator length influence internal pressure dynamics and resultant force outputs remains incomplete. This research addresses this gap by developing and optimizing a state-space model to better predict the dynamic relationships between input air pulses, internal pressure variations, and generated forces across various actuator lengths.

Methods



BPAs of lengths 140 mm, 310 mm, and 600 mm were tested using a rigid 80/20 T-slot aluminum frame and 3D printed parts. Actuation was precisely controlled using Festo VUVG-B14-T32C-AZT-F-1T1L air solenoid valves mounted on a VTUG valve manifold, administering air pulses with durations of 10 ms and 40 ms at frequencies ranging from 8 Hz to 16 Hz. Force output measurements were obtained with a Loadstar RAS1 S-beam force sensor, while two NXP MPX5700AP pressure sensors recorded pressure data before and after the actuator. Pulse trains were continuously applied for 3 seconds, with 5-second intervals for rest. Data collection and system control were facilitated by a Teensy 4.1 microcontroller, with subsequent data processing conducted using MATLAB.

Fig 2. Experimental setup for pulse actuation.

Modeling and Optimization

A state-space representation of the actuator system, conceptualized as interconnected spring-damper-mass systems, was developed to model valve actuation and pressure dynamics. Particle Swarm Optimization (PSO) [6] was applied to refine state-space parameters by minimizing prediction errors between modeled and experimental pressure data.

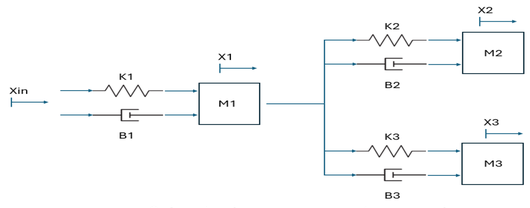


Fig 3. Input valve-pressure system model.

Additionally, a nonlinear model correlating measured pressures to actuator force output was established using trust-region reflective least squares optimization using the following equation.

$$F = \frac{aP\hat{}2}{P+b}$$

where F is the force output, a & b are derived parameters, and P is the pressure in the system.

Results

The state-space model accurately predicted pressure changes in the actuators at different pulse frequencies and lengths. Error rates were around 5% for the 310 mm actuator, 10% for the 600 mm, and 16% for the 140 mm.
The nonlinear pressure-to-force model also performed well, with errors below 3% for all actuator lengths. Longer actuators produced higher forces but responded more slowly, while shorter ones reacted faster but needed higher pulse rates to maintain force.

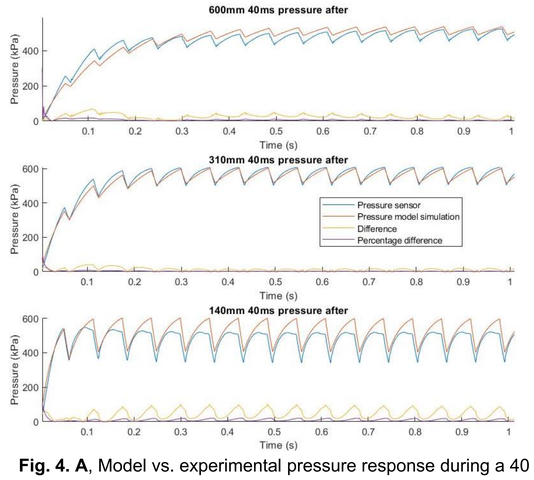
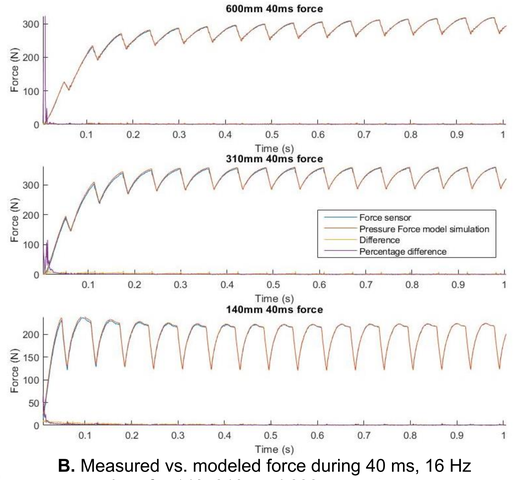


Fig. 4. A, Model vs. experimental pressure response during a 40 ms, 16 Hz pulse train.



B. Measured vs. modeled force during 40 ms, 16 Hz pulses for 140, 310, and 600 mm actuators.

Resultant Variable Values

Valve – Pressure Model Variables			
Length (mm)	600	310	140
M1	1.00e-08	1.00e-08	1.00e-08
M2	0.249	1.00e-08	1.00e-08
M3	1.00e-08	1.13e+03	1.00e-08
K1	2.38e+03	2.58e+04	6.32e+04
K2	0.01	1.93e+03	1.00e+05
K3	0.01	1.42e+03	3.56e+03
B1	2.25	3.35e+03	1.78e+03
B2	2.06	4.31e+03	9.76e+05
B3	0.0031	1.46e+03	1.00e-08

Table 1. Derived results for the variables of the input valve-Pressure model.

Pressure – Force model equations	
Length (mm)	Equations
600	$\frac{0.788 * Pressure^2}{Pressure + 154}$
310	$\frac{0.746 * Pressure^2}{Pressure + 159}$
140	$\frac{0.729 * Pressure^2}{Pressure + 361}$

Table 2. Derived results for the variables of the Pressure-Force model.

Conclusions

- A PSO-optimized state-space model and a nonlinear pressure-to-force model together provided accurate predictions of BPA behavior.
- Actuator length significantly influenced performance: longer actuators produced higher forces with slower responses, while shorter ones responded quickly but required higher pulse rates.
- These insights support the customization of BPAs for targeted soft robotic applications, balancing speed and force requirements.

Acknowledgements

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